

DISASTER CROWDSOURCING EXCHANGE

Crowdsourcing Analysis of Imagery

Join Slack: <http://bit.ly/CrowdExSlack>

Slack Channel # p2-image-analysis

Project Leads: Chris Vaughan (FEMA), George Corea (SBTF), Schuyler Earle

Participant Coordinator(s):

Task Coordinator:

Room (In-Person):

Call / Video (Remote): Project 2: <https://fema.connectsolutions.com/crowdex1>
1-800-320-4330 Pin 153669 (listen only)

GitHub:

Tasks and Outcomes:

- Data Sources: Quantum Spatial, NOAA, CAP
- Tools: [FEMA Imagery Analysis Tool](#), [Esri Image Interpretation tool](#), [Pybossa](#)
- Enhance the FEMA Imagery Analysis tool to expand the capabilities
 - Track changes to a data point over time, identify building material type, mark road type and status, distinguish commercial and residential etc
- Explore using the crowdsourced data for machine and deep learning

IP

1. *Each project chooses its own license.*

The project's leaders choose a license for it - that could be AGPL, or MIT, or NPOSL, as they like. The choice of whether to restrict commercial activities rests with project leaders.

2. *Each project has a CTA.*

The copyright of any contributions to the project gets assigned to the project's leaders. The leaders can choose who specifically to assign it to, but individual developers don't keep their copyright.

Outcomes:

- Prototypes of an enhanced Imagery Analysis Tools
- Prototype Applying Deep and Machine Learning capabilities to automate Satellite and Other Images for classification and damage assessment

Preparation Work:

- Need standards and guidance to build into this tool for different audiences (CAP, SBTF, middle school, etc.)
- Try to tap into education as much as possible to increase volunteers
- Consider how this could be used to make products for state and local
- Snip portions of satellite imagery and create a stand alone offline map people can use on the ground offline
- Consider how different images are used for different reasons, find a tool that allows you to easily bring together all imagery (GeoQ)

Participants: [add your name and email so we have a record of who participated]

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SBTF DRAFT Requirements for activations -if SBTF was designing this workflow the following would be our key goals.

- Several people should be able to use it at the same time (we can have dozens of volunteers working at the same time)
 - Avoid duplication of efforts
 - No anonymous entries.
 - Ideally, the system should allow data entry only by volunteers that have been approved to work on the project (so that if prior training is needed, they don't start to add data skipping the instructions)
- Need a system of data validation. Not only for data format, but also a way to analyze the quality of the data entered. For instance, if someone is marking all the roads that are blocked as "open", the deployment leads need to be able to address the issue in some way - now if we had several entries with this problem we were unpublishing those entries (selecting "no" in the published column), fix the problem and publish them again.

- Data should be available on a spreadsheet.
- The process should be easy to understand for people without a GIS background.
- Data analysis to identify information gaps and data quality (identify mistakes) shouldn't be too complex.

Link to SBTF data sample (names of the volunteers that have entered the data have been anonymized for privacy):

https://docs.google.com/spreadsheets/d/14_894yDZemeC5Yvq7xhbdImzptVa62VZ-AW4TxppWbo/edit?usp=sharing